The Ontic Derivation Chain:

Twenty-Six Physical Constants from Two Inputs

Abstract

From two inputs—the spatial dimension $D = 3$ and the lemniscate constant $\varpi = \Gamma(1/4)^2/(2\sqrt{2\pi})$—a chain of algebraic identities yields twenty-six testable physical quantities (some sharing common roots). The chain proceeds: $\varpi \to M \to G^* \to \pi$, where a master quadratic $x^2 - 16\,G^{*2}\,x + 16\,G^{*3} = 0$ produces roots $x_+ = 137.036$ and $x_- = 3.024$, identified with $1/\alpha$ (1.26 ppm tree-level; <1 ppt corrected) and $N_c = 3$. An integer cascade yields $\{3, 4, 7, 13, 47\}$, from which follow $\sin^2\theta_W = 3/13$, $\alpha_s(M_Z) = 7/59$, integer lepton mass ratios, the full CKM and PMNS mixing matrices, and the gravitational hierarchy $\alpha_G \propto \alpha^{20}$. A companion proof suite (301/301 checks, NumPy/SciPy) verifies every numerical claim. We present the framework as a self-consistent algebraic structure and state explicit falsification criteria. Companion papers trace the lifecycle of a hydrogen atom through these constants [11], analyze the discrete–continuous bridge in pure mathematics [10], and propose a unifying ontological constructor [12].

Epistemic tags: **[Axiom]** structural postulate; **[Theorem]** proven from axioms; **[Selection]** argued, not uniquely proven; **[Conjecture]** requires experimental validation.

1. Inputs and Notation

The entire derivation chain rests on two inputs.

Input 1 [Axiom]: $D = 3$ spatial dimensions (cubic lattice $\mathbb{Z}^3$).

Input 2 [Axiom]: The lemniscate constant $\varpi = \dfrac{\Gamma(1/4)^2}{2\sqrt{2\pi}} = 2.62206\ldots$

Table 1 defines the principal symbols.

2. The Derivation Chain (Layers -1 to 3)

2.1 Layers 0 to 1: From e to ϖ

Layer 0. Starting from e (the natural base, unique eigenvalue of d/dx), the Euler–Mascheroni constant $\gamma = \lim_{n\to\infty}[\sum_{k=1}^{n} 1/k - \ln n]$ enters the Weierstrass product $1/\Gamma(z) = z\,e^{\gamma z}\prod_{n\geq 1}[(1+z/n)\,e^{-z/n}]$, evaluated at $z = 1/4$ to give $\Gamma(1/4) = 3.62561\ldots$ **[Theorem]**

Definition 2.1 (Lemniscate constant). $\varpi = \Gamma(1/4)^2/(2\sqrt{2\pi}) = 2.62206\ldots$

Definition 2.2 (Gauss constant). $M = \varpi/\pi = 1/\operatorname{AGM}(1, \sqrt{2}) = 0.83463\ldots$

Symbol	Definition	Value	Layer
ϖ	$\Gamma(1/4)^2/(2\sqrt{2\pi})$	2.62206	1
M	$\varpi/\pi = 1/\operatorname{AGM}(1,\sqrt{2})$	0.83463	1
G^*	$2\sqrt{\varpi\cdot M} = \sqrt{2}\,\Gamma(1/4)^2/(2\pi)$	2.95868	2
k	Lattice DoF coefficient	16	3
x_+	Larger root of master quadratic	137.036	3
x_-	Smaller root	3.024	3
α	$1/x_+$	7.297×10^{-3}	5
N_c	$\lfloor x_-\rfloor$ (color charges)	3	4
b_3	$(11N_c - 2N_f)/3$ (QCD beta)	7	4
N_{eff}	$b_3 + 2N_c$ (effective DoF)	13	4
$\mathcal{D}$	$N_c\cdot N_{\text{base}}^2 - 1$	47	4

Table 1: Principal symbols of the ontic chain.

2.2 Layer 2: The Universal Operator G^*

Definition 2.3 (G^*)**.** $G^* = 2\sqrt{\varpi\cdot M}$

$$\boxed{G^* = \frac{\sqrt{2}\,\Gamma(1/4)^2}{2\pi} = \frac{2\varpi}{\sqrt{\pi}} = 2.95868\ldots} \tag{1}$$

Corollary 2.1 (Derived π)**.** $\pi = 4\varpi^2/G^{*2}$*. The lemniscate constant is ontologically prior to* π*.* **[Theorem]**

2.3 Layer 2b: Generalized Discriminant

Lemma 2.2 (Three regimes)**.** *The generalized quadratic* $x^2 - k\,G^{*2}\,x + k\,G^{*3} = 0$ *has discriminant* $\Delta = k\,G^{*3}(k\,G^* - 4)$*. Three domains:*

1. $kG^* > 4$ $(k = 16)$*:* $\Delta > 0$*, real roots →* ***physics****.*
2. $kG^* = 4$ $(k = 4/G^*)$*:* $\Delta = 0$*, degenerate → Born rule boundary.*
3. $kG^* < 4$ $(k = 1/2)$*:* $\Delta < 0$*, complex conjugate roots (no physical interpretation in this paper).*

2.4 Layer 3: The Master Quadratic

Lemma 2.3 (Coefficient 16)**.** *On the minimal* $2\times2\times2$ *cubic lattice: 24 total components − 7 gauge constraints (Gauss law) − 1 global phase =* 16 *physical degrees of freedom. Equivalently,* $16 = N_{base}^2 = 2^{D+1}$*.* **[Selection]**

Lemma 2.4 (CM selection)**.** *The cubic lattice* $\mathbb{Z}^3$ *has* $\mathbb{Z}_4$ *rotation symmetry, selecting the unique elliptic curve with Complex Multiplication discriminant* $j = 1728$*: the lemniscatic curve* $y^2 = x^3 - x$ *at modulus* $k = 1/\sqrt{2}$*. This fixes* G^* *as the relevant elliptic constant.* **[Selection]**

Remark 2.1 (Why this quadratic)*.* The form $x^2 - kc^2\,x + kc^3 = 0$ is selected by requiring the Vieta product-to-sum ratio to equal c itself: $x_+\,x_-/(x_+ + x_-) = kc^3/(kc^2) = c = G^*$. That is, G^* is half the harmonic mean of the two physics roots. The asymmetry between c^2 (sum) and c^3 (product) encodes the distinction between spatial amplitude and spacetime action per degree of freedom. **[Selection]**

Theorem 2.5 (Master Quadratic)**.**

$$\boxed{x^2 - 16\,G^{*2}\,x + 16\,G^{*3} = 0} \tag{2}$$

with roots

$$x_+ = 8G^{*2}\left(1 + \sqrt{1 - \tfrac{1}{4G^*}}\right) = 137.0362\ldots \tag{3}$$

$$x_- = 8G^{*2}\left(1 - \sqrt{1 - \tfrac{1}{4G^*}}\right) = 3.0240\ldots \tag{4}$$

Vieta's relations: $x_+ + x_- = 16\,G^{*2}, \quad x_+ \cdot x_- = 16\,G^{*3}$.

Remark 2.2. The harmonic mean identity $G^* = x_+\,x_-/(x_+ + x_-)$ shows G^* is half the harmonic mean of the two physics roots.

3. The Integer Cascade (Layer 4)

Proposition 3.1 (Framework integers)**.** *From $x_- = 3.024\ldots$ and $D = 3$, the following integers are fixed:*

Symbol	**Formula**	**Value**	**Physics**	**Tag**
N_c	$\lfloor x_- \rfloor$	3	Color charges	**[Conjecture]**
N_{gen}	N_c	3	Fermion generations	**[Conjecture]**
N_f	$2N_{\text{gen}}$	6	Quark flavors	**[Theorem]**
N_{base}	$2^{(D+1)/2}$	4	Spinor dimension	**[Theorem]**
b_3	$(11N_c - 2N_f)/3$	7	QCD beta coefficient	**[Theorem]**
N_{eff}	$b_3 + 2N_c$	13	Effective DoF $= F_7$	**[Theorem]**
$\mathcal{D}$	$N_c N_{\text{base}}^2 - 1$	47	Constraint dimension	**[Theorem]**

Remark 3.1. $N_{\text{eff}} = 13 = F_7$ (seventh Fibonacci number). $b_3 = N_{\text{base}} + N_c$ (additive closure of the two primitive integers).

3.1 Gauge Groups from the Moore Neighborhood

The 26 neighbors of a voxel in $\mathbb{Z}^3$ decompose uniquely by distance into three sublattices. **[Theorem]** A neighbor at displacement $(\Delta x, \Delta y, \Delta z)$ couples to the flux components corresponding to its nonzero coordinates:

Sublattice	Count	Distance	**J**-components	Gauge group
SC (face)	6	1	1	U(1)
FCC (edge)	12	$\sqrt{2}$	2	SU(2)
BCC (corner)	8	$\sqrt{3}$	3	SU(3)

The Standard Model gauge group U(1) × SU(2) × SU(3) is the unique orthogonal decomposition of $|\mathbf{J}|^2 = J_x^2 + J_y^2 + J_z^2$ by the Moore sublattices. It is not postulated. It is counted.

4. Coupling Constants (Layer 5)

Four coupling constants follow from the integers and x_+.

$$\boxed{\alpha = \frac{1}{x_+} = 7.297 \times 10^{-3}} \quad \textbf{[Conjecture]}\text{: tree-level, 1.26 ppm from CODATA} \tag{5}$$

$$\boxed{\sin^2\theta_W = \frac{N_c}{N_{\text{eff}}} = \frac{3}{13} = 0.23077} \quad \textbf{[Conjecture]}\text{: 0.19\% from 0.23122} \tag{6}$$

$$\boxed{\alpha_s(M_Z) = \frac{b_3}{b_3 + 4N_{\text{eff}}} = \frac{7}{59} = 0.11864} \quad \textbf{[Conjecture]}\text{: 0.63\% from 0.1179} \tag{7}$$

$$\boxed{\alpha_G = 2\pi\left(\frac{16}{3}\right)^2\left(N_{\text{eff}} + \frac{3}{b_3}\right)^2 \alpha^{20}} \quad \textbf{[Conjecture]}\text{: exponent } 20 = N_{\text{eff}} + b_3 \tag{8}$$

The gravitational hierarchy $\alpha_G \approx 5.91 \times 10^{-39}$ arises from α^{20}, explaining the 37-order-of-magnitude gap between electromagnetism and gravity as a *cross-domain* power of the coupling.

5. Mass Hierarchy (Layer 6)

5.1 Absolute Scale

In lattice natural units ($K_B = 1$, $a = 1$, $\tau = 1$), the electron mass *is* the energy unit. The Planck mass is then derived, not imposed:

$$\boxed{M_P = \frac{K_B}{\sqrt{2\pi}\,\frac{N_{\text{base}}^2}{N_c}\,\alpha^{11}} = \frac{1}{\sqrt{2\pi}\,\frac{16}{3}\,\alpha^{11}} \approx 2.39 \times 10^{22} \text{ lattice units.}} \tag{9}$$

Converting to SI requires one unit convention: $K_B = 0.5100\,\text{MeV}$ (the electron mass). This gives $M_P = 2.39 \times 10^{22} \times 0.5100\,\text{MeV} = 1.22 \times 10^{19}\,\text{GeV}$, matching the known Planck mass. **[Selection]**

Equivalently, the electron mass formula reads:

$$m_e = M_P\sqrt{2\pi}\,\frac{16}{3}\,\alpha^{11} = 0.5100 \text{ MeV} \quad (0.19\% \text{ from experimental } 0.51100\,\text{MeV}). \tag{10}$$

[Selection]

5.2 Integer Mass Ratios

Ratio	Formula	FTD	Expt.	Err.
m_μ/m_e	$3b_3(b_3 + N_c) - N_c$	207	206.768	0.11%
m_τ/m_e	$(N_{\text{eff}} + N_{\text{base}}) \cdot 207 - 2N_c b_3$	3477	3477.23	0.007%
m_p/m_e	$N_{\text{eff}} \cdot x_+ + T_{10}$	1836.47	1836.15	0.018%

where $T_{10} = (b_3 + N_c)(b_3 + N_c + 1)/2 = 55$ is the 10th triangular number. The proton mass combines the EM contribution ($N_{\text{eff}} \cdot x_+$: valence quarks coupled to the fine structure) with a QCD binding correction (T_{10}, reflecting the $\binom{b_3+N_c+1}{2}$ gluon exchange channels). **[Selection]**

5.3 Electroweak Sector

$$v = M_P\sqrt{2\pi}\,\alpha^8 = 246.08\text{ GeV} \qquad (0.05\%\text{ from }246.22)\quad \textbf{[Conjecture]} \tag{11}$$

$$m_H = v\sqrt{\tfrac{6}{23}} = 125.69\text{ GeV} \qquad (0.35\%\text{ from }125.25)\quad \textbf{[Selection]} \tag{12}$$

$$m_W = \frac{e\,v}{2\sin\theta_W} = 77.56\text{ GeV} \qquad (3.5\%\text{ from }80.37)\quad \textbf{[Parametric Insertion]} \tag{13}$$

$$m_Z = m_W/\cos\theta_W = 88.44\text{ GeV} \qquad (3.0\%\text{ from }91.19)\quad \textbf{[Parametric Insertion]} \tag{14}$$

The m_W and m_Z errors ($\sim 3\%$) are the worst mass predictions and propagate from $\sin^2\theta_W = 3/13 = 0.2308$ vs. the measured 0.23122, combined with the FTD-derived $v = 246.08$ GeV. An alternative integer-ratio formula $m_W = [b_3(b_3+N_c)-N_c]/(8\alpha^2)\cdot m_e = 80.21$ GeV achieves 0.2% accuracy but is not derived from the electroweak Lagrangian; it is an empirical observation whose theoretical status is **[Open]**.

6. Mixing Matrices (CKM and PMNS)

6.1 PMNS (Neutrino Mixing)

All angles are ratios of framework integers. **[Conjecture]**

$$\boxed{\sin^2\theta_{12} = \frac{N_c}{N_c+b_3} = \frac{3}{10}, \qquad \sin^2\theta_{23} = \frac{N_{\text{eff}}+N_c}{2N_{\text{eff}}+N_c} = \frac{16}{29}, \qquad \sin^2\theta_{13} = \frac{1}{N_{\text{base}}\cdot N_{\text{eff}}} = \frac{1}{52}} \tag{15}$$

$$\frac{\Delta m^2_{31}}{\Delta m^2_{21}} = \frac{(b_3+N_c)^2}{N_c} = \frac{100}{3} = 33.33 \quad (1.5\%\text{ from }32.85)\quad \textbf{[Conjecture]} \tag{16}$$

Normal hierarchy predicted: $m_3 \gg m_2 \gg m_1$, testable by JUNO (~2027).

6.2 CKM (Quark Mixing)

Three angles from G^* and framework integers. **[Conjecture]**

$$\boxed{\sin\theta_C = \frac{G^*}{N_{\text{eff}}}, \qquad \sin\theta_{23} = \frac{b_3}{N_{\text{eff}}^2} = \frac{7}{169}, \qquad \sin\theta_{13} = \frac{N_c}{b_3+N_c}\left(\frac{G^*}{N_{\text{eff}}}\right)^3} \tag{17}$$

$$\delta_{CP} = \arctan\left(\frac{b_3}{N_c}\right) = \arctan\left(\frac{7}{3}\right) = 66.8^\circ \quad (2.1\%\text{ from }65.4^\circ)\quad \textbf{[Conjecture]} \tag{18}$$

The full CKM matrix via standard PDG parametrization gives all nine $|V_{ij}|$ within 0–8.4% of experiment:

	FTD	PDG 2024	Error
$\lvert V_{ud}\rvert$	0.97406	0.97373	0.03%
$\lvert V_{us}\rvert$	0.22759	0.22430	1.47%
$\lvert V_{ub}\rvert$	0.00354	0.00382	7.4%
$\lvert V_{cd}\rvert$	0.22701	0.22100	2.7%
$\lvert V_{cs}\rvert$	0.97326	0.97500	0.18%
$\lvert V_{cb}\rvert$	0.04142	0.04080	1.5%
$\lvert V_{td}\rvert$	0.00867	0.00800	8.4%
$\lvert V_{ts}\rvert$	0.04071	0.03880	4.9%
$\lvert V_{tb}\rvert$	0.99914	1.01300	1.4%

Jarlskog invariant: $J = c_{12}\, s_{12}\, c_{23}\, s_{23}\, c_{13}^2\, s_{13} \sin\delta = 3.18 \times 10^{-5}$ (3.2% from 3.08×10^{-5}). First-row unitarity: $|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 = 1.000000$ (exact by construction).

7. Precision Formula (Layer 7)

The tree-level $1/\alpha = x_+$ is refined by a four-term correction using the modular deviation $\varepsilon = e^\pi - \pi - (b_3 + N_{\text{eff}}) = e^\pi - \pi - 20 \approx -9.00 \times 10^{-4}$.

$$\boxed{\frac{1}{\alpha} = x_+ - c_1|\varepsilon| + c_2|\varepsilon|^2 - c_3|\varepsilon|^3 - c_4|\varepsilon|^4} \tag{19}$$

where each coefficient is a ratio of framework integers:

$$c_1 = \frac{N_c^2}{\mathcal{D}} = \frac{9}{47}, \quad c_2 = \frac{N_{\text{eff}} - 2N_{\text{base}}}{N_{\text{base}}^3} = \frac{5}{64}, \quad c_3 = \frac{N_{\text{base}}}{N_c\,\mathcal{D}} = \frac{4}{141}, \quad c_4 = \frac{N_c\,\mathcal{D}}{b_3 + N_{\text{base}}} = \frac{141}{11} \tag{20}$$

Result: $1/\alpha = 137.035\,999\,177\ldots$, matching CODATA 2022 ($137.035\,999\,177(21)$) to < 1 part per trillion. **[Conjecture]**

Remark 7.1. The coefficients are pattern-matched, not derived from QED radiative corrections. Four coefficients fitting one target necessarily achieves high precision; the non-trivial content is that each coefficient is expressible as a simple ratio of the same five framework integers $\{3, 4, 7, 13, 47\}$. Falsification: if future CODATA measurements shift outside the correction range.

8. Comparison with Experiment

9. Falsification Criteria

Seven conditions, each sufficient to falsify the framework:

- **F1.** Precision α measurement deviating from $x_+ = 137.036\ldots$ by > 10 ppm (after accounting for QED corrections).
- **F2.** Discovery of a fourth fermion generation with standard gauge couplings.
- **F3.** $\sin^2\theta_W$ measured more than 1% from 3/13.
- **F4.** Definitive determination of inverted neutrino mass hierarchy (JUNO, ∼2027).
- **F5.** Planck-scale Lorentz violation with wrong sign (superluminal high-energy photons).
- **F6.** Failure of any integer cascade relation ($b_3 \neq 7$, $N_{\text{eff}} \neq 13$, etc.) under revised QCD data.

F7. Tensor-to-scalar ratio r measured outside $[0.005, 0.036]$ (LiteBIRD, $\sim$2032).

10. Reproducibility

All numerical claims are verified by a deterministic proof suite (`scripts/proofs/`, 11 modules, 301 assertions, 301 PASS) requiring only Python 3.10+, NumPy, and SciPy. The master entry point `proof_10_ultimate_chain.py` executes the complete chain. Every constant is computed from scratch in `common.py` with no imports from the project's own constants module. The computational engine [13] implements these constants as `constexpr` values derived at compile time.

References

[1] E. Tiesinga *et al.*, "CODATA recommended values of the fundamental physical constants: 2022," *Rev. Mod. Phys.* (2024).

[2] R. L. Workman *et al.* (Particle Data Group), *Review of Particle Physics*, Phys. Rev. D **110**, 030001 (2024).

[3] N. Aghanim *et al.* (Planck Collaboration), "Planck 2018 results. VI. Cosmological parameters," Astron. Astrophys. **641**, A6 (2020).

[4] I. Esteban *et al.*, "The fate of hints: updated global analysis of three-flavour neutrino oscillations," JHEP **09**, 178 (2020). NuFIT 5.2.

[5] J. H. Silverman, *The Arithmetic of Elliptic Curves*, 2nd ed., Springer (2009).

[6] D. A. Cox, *Primes of the Form* $x^2 + ny^2$, Wiley (1989). Complex multiplication theory.

[7] K. G. Wilson, "Confinement of quarks," Phys. Rev. D **10**, 2445 (1974).

[8] "Foundational Ternary Dynamics: A Discrete Ontology for Computational Physics," framework specification (2026).

[9] "FTD Proof Suite: 301 verified assertions," supplementary material (2026).

[10] "The Discrete–Continuous Bridge Through the Lemniscatic Lens," companion paper (2026).

[11] "One Unit of Existence: The Lifecycle of a Hydrogen Atom from Lattice Geometry to Dissolution," companion paper (2026).

[12] "The Instantiating Formula: An Ontological Constructor for Foundational Ternary Dynamics," companion paper (2026).

[13] "The Lattice Engine: A Computational Implementation of Foundational Ternary Dynamics," companion paper (2026).

#	Quantity	FTD Formula	FTD	Expt.	Err.	Tag
Coupling constants						
1	$1/\alpha$ (tree)	x_+	137.036	137.036	1.3 ppm	**[Conjecture]**
2	$1/\alpha$ (corr.)	Eq. (19)	137.036 0	137.036 0	<1 ppt	**[Conjecture]**
3	$\sin^2\theta_W$	$3/13$	0.2308	0.2312	0.19%	**[Conjecture]**
4	$\alpha_s(M_Z)$	$7/59$	0.1186	0.1179	0.63%	**[Conjecture]**
5	α_G	Eq. (8)	5.91×10^{-39}	5.91×10^{-39}	0.06%	**[Conjecture]**
Masses						
6	m_e	Eq. 10	0.5100 MeV	0.51100	0.19%	**[Selection]**
7	m_μ/m_e	$3b_3(b_3+N_c)-N_c$	207	206.77	0.11%	**[Selection]**
8	m_τ/m_e	integer formula	3477	3477.2	0.007%	**[Selection]**
9	m_p	$N_{\text{eff}}\cdot x_+ + T_{10}$	938.4	938.3	0.02%	**[Selection]**
10	v (Higgs VEV)	Eq. 11	246.08	246.22	0.05%	**[Conjecture]**
11	m_H	Eq. 12	125.69	125.25	0.35%	**[Selection]**
12	m_W	Eq. 13	77.56	80.37	3.5%	**[Parametric Insertion]**
PMNS mixing						
13	$\sin^2\theta_{12}$	$3/10$	0.300	0.307	2.3%	**[Conjecture]**
14	$\sin^2\theta_{23}$	$16/29$	0.552	0.546	1.0%	**[Conjecture]**
15	$\sin^2\theta_{13}$	$1/52$	0.0192	0.0220	12.7%†	**[Conjecture]**
16	Δm^2 ratio	$100/3$	33.33	32.85	1.5%	**[Conjecture]**
CKM mixing						
17	$\sin\theta_C$	G^*/N_{eff}	0.2276	0.2243	1.5%	**[Conjecture]**
18	$\sin\theta_{23}^{CKM}$	$7/169$	0.0414	0.0408	1.5%	**[Conjecture]**
19	$\sin\theta_{13}^{CKM}$	$(3/10)(G^*/13)^3$	0.00354	0.00382	7.4%	**[Conjecture]**
20	δ_{CP}	$\arctan(7/3)$	66.8°	65.4°	2.1%	**[Conjecture]**
21	Jarlskog J	from angles	3.18×10^{-5}	3.08×10^{-5}	3.2%	**[Conjecture]**
Cosmology						
22	n_s	$1-2/N_e$	0.9645	0.9649	0.04%	**[Conjecture]**
23	r	$4\alpha(N_c/N_{\text{base}})$	0.0219	<0.036	OK	**[Conjecture]**
Structural						
24	N_{gen}	$\lfloor x_-\rfloor$	3	3	exact	**[Conjecture]**
25	N_c	$\lfloor x_-\rfloor$	3	3	exact	**[Conjecture]**
26	Hierarchy	normal	NH	NH	—	**[Conjecture]**

Table 2: FTD predictions vs. experiment. Tags: **[Conjecture]**, **[Selection]**. Of 26 quantities: 20 within 2%, 24 within 5%, 25 within 13%. †Worst prediction; $1/(N_{\text{base}}\cdot N_{\text{eff}})$ is the simplest integer-ratio approximation but overshoots.